

Block Storage

1. Overview

Block storage stores data as **fixed-size chunks called blocks**, where each block has a unique address/identifier.

Instead of storing data as files or rows, the storage device exposes raw blocks that another system (OS, database, filesystem) can manage.

Example concept:

None

Block 001

Block 002

Block 003

...

Applications usually do **not** access blocks directly. Operating systems and databases do.

2. Simple Analogy

Think of a hard disk divided into equal sectors:

None

Disk = many small numbered boxes

Each box stores bytes

You read/write specific boxes.

3. Key Characteristics

Fixed-size units

Random read/write access

High performance

Low-level storage abstraction

Used beneath filesystems/databases

4. How It Works

A block device gives access like:

None

Read block #1024

Write block #2048

Then higher layers build:

- File systems
- Databases
- Virtual machine disks

5. Common Examples

- SSD drives
- HDD drives
- SAN volumes
- Cloud volumes:
 - Amazon Web Services Amazon EBS
 - Google Cloud Persistent Disk
 - Microsoft Azure Managed Disks

6. File Storage vs Block Storage

Feature	File Storage	Block Storage
Interface	Files/folders	Raw blocks
Human readable	Yes	No
Needs filesystem	Already included	Yes
Random I/O	Good	Excellent

Typical use	Docs/media	DB disks / VM disks
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7. Object Storage vs Block Storage

Feature	Object Storage	Block Storage
Stores	Objects/files	Blocks
Access	API / HTTP	Mounted disk
Latency	Higher	Lower
Scale	Massive	Strong but attached style
Best for	Images/backups	Databases/OS disks

8. Why Important in System Design

Even if users don't mention it directly, many systems depend on block storage:

Databases

Database pages stored on block devices.

Virtual Machines

OS disk runs on block storage.

High IOPS workloads

Transaction systems need fast random writes.

9. Example: SQL Database

None

MySQL/PostgreSQL



Filesystem



Block Storage (SSD/EBS)

Good block storage improves DB latency.

10. Example: Cloud Deployment

None

EC2 instance

+ attached EBS volume

Application runs on VM, data on persistent block volume.

11. Why Rarely Mentioned in Web App HLD

In many web application interviews, you choose:

- SQL DB
- NoSQL DB
- Cache
- CDN
- Queue

Underlying block storage is abstracted by cloud providers.

So it matters more in:

- Infra engineering
- Database engineering
- Storage systems
- Performance tuning

12. Performance Benefits

Block storage is strong for:

- Random IOPS
- Low latency
- Transaction logs
- Frequent updates
- Database indexes

13. Common Concepts

IOPS

Input/output operations per second.

Throughput

MB/s transferred.

Latency

Time per read/write.

14. Durability Features

Often includes:

- Replication
- Snapshots
- RAID
- Multi-zone copies (provider dependent)

15. Trade-offs

Requires filesystem or DB layer

Harder to scale than object storage

Usually attached to instances

More expensive for large cold storage

16. Interview Answer Example

Q: Where should production PostgreSQL data live?

I'd place PostgreSQL data on high-performance SSD-backed block storage for low-latency random I/O and durable persistence.

17. Architecture View

None

User Request

→ App Server

→ Database

→ Filesystem

→ Block Storage

18. Short Revision Notes

None

Block Storage:

- Fixed-size chunks
- Raw disk-like storage
- High performance
- Used by DBs / VMs

Examples:

EBS, SSD, SAN

19. One-Line Summary

Block storage provides raw fixed-size storage blocks with fast random access, making it foundational for databases, filesystems, and virtual machine disks rather than direct web application data access.